Prime periods of the chocolate numbers
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Abstract

For every prime $p \equiv 2, 3 \pmod 5$ with $p \neq 2$, the chocolate numbers have eventual period $p(p-1)$ modulo p. More precisely, the shift $p(p-1)/2$ multiplies the sequence by the quadratic character of -5, starting at index $(p+1)/2$. The proof uses integer divided powers, so no factorial is inverted in characteristic p.

Task 1: the source question and its resolution

Let B_n count the ways of breaking a 2 by n chocolate bar into unit squares, with the order of the breaks recorded. The source [1] gives $B_1 = 1$ and, for $n \geq 2$,

$$B_n = (2n-2)! + \sum_{i=1}^{n-1} \binom{2n-2}{2i-1} B_i B_{n-i}. \tag{1}$$

Its Conjecture 23 (Conjecture 3 in the preprint) asks for an eventual period divisible by $p(p-1)$ when $p \neq 2, 5$ and $p \not\equiv \pm 1 \pmod 5$. The following identifies $p(p-1)$ itself as such a period.

Theorem 1. *Let p be an odd prime for which 5 is a quadratic nonresidue. Put $T = p(p-1)/2$ and $\epsilon = \left(\frac{-5}{p}\right) \in \{1, -1\}$. For every $n \geq (p+1)/2$,*

$$B_{n+T} \equiv \epsilon B_n \pmod p. \tag{2}$$

Consequently $B_{n+p(p-1)} \equiv B_n \pmod p$ throughout this range.

Quadratic reciprocity gives $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, so these are exactly the primes in the source question. The theorem specifies a period, without asserting that it is the least period. For example, the source already proves period 3 modulo 3 for $n \geq 2$; our formula recovers it with $T = 3$.

Divided powers and the auxiliary series

For a commutative ring R, let $\mathcal{H}(R)$ be the ring of sequences $A = (a_0, a_1, \ldots)$, written $A = \sum_{j \geq 0} a_j t^{[j]}$, with product

$$(AB)_j = \sum_{i=0}^{j} \binom{j}{i} a_i b_{j-i}.$$

Over $\mathbb{Q}$, $t^{[j]}$ is $t^j/j!$. The operator D is the shift $(DA)_j = a_{j+1}$. Pascal's identity proves that D is a derivation. A series with constant coefficient 1 has a unique inverse, recursively in the same coefficient ring.

Define integers d_j by

$$d_0 = 1, \qquad d_1 = 0, \qquad d_{j+2} = (j^2 - j - 1)d_j \quad (j \geq 0), \tag{3}$$

and put $U = \sum d_j t^{[j]} \in \mathcal{H}(\mathbb{Z})$. All odd d_j vanish. Writing $t = t^{[1]}$, recurrence (3) is equivalent to

$$(1 - t^2)D^2U + U = 0.$$

Indeed, the jth coefficient of t^2D^2U is $j(j-1)d_j$. Let $H = (1-t^2)^{-1}$; its coefficient is $j!$ for even j and zero for odd j. Set $Y = -(DU)U^{-1}$. Then $Y_0 = 0$ and

$$DY = H + Y^2. \tag{4}$$

The coefficients of (4) determine Y uniquely from Y_0. Comparing them with (1), including the zero even coefficients, gives the identity

$$Y = \sum_{n \geq 1} B_n t^{[2n-1]} = -(DU)U^{-1} \quad \text{in } \mathcal{H}(\mathbb{Z}). \tag{5}$$

This is the source's auxiliary logarithmic derivative in integral form. All these identities may therefore be reduced coefficientwise modulo p.

Proof of Theorem 1

We now work in $\mathcal{H}(\mathbb{F}_p)$. Iterating (3) p times and using that $j, j+2, \ldots, j+2(p-1)$ run through $\mathbb{F}_p$ yields

$$d_{j+2p} = d_j \prod_{r \in \mathbb{F}_p} (r^2 - r - 1).$$

Let $\alpha, \beta \in \mathbb{F}_{p^2}$ be the roots of $X^2 - X - 1$. The nonresidue hypothesis implies $\alpha^p = \beta$ and $\beta^p = \alpha$. Since $\prod_{r \in \mathbb{F}_p}(X - r) = X^p - X$, we obtain

$$\prod_r (r^2 - r - 1) = (\alpha^p - \alpha)(\beta^p - \beta) = -(\alpha - \beta)^2 = -5.$$

Thus, on putting $\delta = D^p$,

$$\delta^2 U = -5U, \qquad \delta^p U = (-5)^{(p-1)/2}\delta U = \epsilon \delta U. \tag{6}$$

The last equality is Euler's criterion; $p \neq 5$ ensures $\epsilon^2 = 1$.

For every derivation in characteristic p, its pth iterate is also a derivation: in the iterated Leibniz rule all intermediate binomial coefficients $\binom{p}{i}$ vanish. Hence both D^p and D^{p^2}, and therefore

$$L = D^{p^2} - \epsilon D^p,$$

are derivations. Equation (6) gives $LU = 0$. As L commutes with D, it also gives $L(DU) = 0$. Applying L to $UU^{-1} = 1$ gives $L(U^{-1}) = -U^{-2}LU = 0$. Identity (5) now implies $LY = 0$.

Writing y_j for the coefficients of Y, we have proved

$$y_{r+p^2} = \epsilon y_{r+p} \qquad (r \geq 0).$$

Choose $r = 2n - 1 - p$, which is nonnegative exactly when $n \geq (p+1)/2$. Both subscripts are odd, and (5) turns the last identity into (2). Applying it twice proves the stated period because $\epsilon^2 = 1$. This proves the theorem for every admissible prime. $\square$

Proposed follow-up tasks

2. Determine the least eventual period modulo each admissible prime, including a criterion for when it is smaller than the period above.

3. Determine the least starting index for (2), and characterize the primes for which the bound $(p+1)/2$ is optimal.

4. Establish explicit eventual periods modulo p^e for $e \geq 2$, including the dependence on e and on the residue class of p.

5. For $c \in \mathbb{Z}$, replace (3) by $d_{j+2} = (j^2 - j - c)d_j$ and classify the prime periods of the coefficients of $-(DU)U^{-1}$ in terms of $1 + 4c$.

Verification and scope. The proof is uniform in p and n. The accompanying integer program checks independent recurrence coefficients and sample congruences; these finite checks do not replace the proof. All 16 pages of the current preprint and all 17 pages of the final article were read. The dated literature audit records the checked public sources and does not claim exhaustive knowledge of all prior work. Review here is author-agent self-review, not independent peer review.

References

[1] C. Ji, T. Khovanova, R. Park and A. Song, *Chocolate Numbers*, Journal of Integer Sequences **19** (2016), Article 16.1.7. Final article; arXiv:1509.06093v1.